

Jiwoo Kim

PH.D. CANDIDATE · QUANTITATIVE METHODS IN PSYCHOLOGY

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Education

Cornell University

Ithaca, U.S.

PH.D. CANDIDATE IN PSYCHOLOGY

08. 2023 - current

- Major in Psychology, Minor in Applied Statistics
- Advisor: Dr. Felix Thoemmes
- Dissertation topic: Causal Inference with Missing and Carelessly Generated Data

Seoul National University

Seoul, Republic of Korea

MA IN EDUCATION

03. 2020 - 02. 2022

- Major in Educational Measurement and Evaluation
- Advisor: Dr. Sun-Geun Baek
- GPA: 4.3/4.3
- Dean's list for every semester (four times)
- Master's thesis title: The Effect of Self-Directedness on College Adjustment through Academic Resilience

Ewha Womans University

Seoul, Republic of Korea

BA IN EDUCATION AND PSYCHOLOGY

03. 2016 - 02. 2020

- Major in Education and Psychology
- Undergrad Research Advisor: Dr. Kyoung-Hye Seo
- GPA: 4.13/4.3 Summa cum laude
- Dean's list for every semester (eight times)

Research Interests

CAUSAL INFERENCE

causal mediation analysis; causal graphical model; missing data; careless responding; quasi-experimental

STRUCTURAL EQUATION MODEL

latent class analysis; factor analysis

Publications

PUBLISHED

Kim, J., & Thoemmes, F. J. 2026. Treatment effects. *Encyclopedia of Measurement in Social Sciences* (2nd ed.). <https://doi.org/10.1016/B978-0-443-26629-4.00243-4>

Kim, J., & Thoemmes, F. J. 2025. Causal assumptions of the two-wave longitudinal mediation model. *Multivariate Behavioral Research*. <https://doi-org/10.1080/00273171.2025.2443360> (Abstract)

Kim, J., Baek, J. W., Lee, H. S(co-first author)., & Choi, Y. J(second author). 2025. Development and validation of the teacher competency diagnostic scale in online learning environment for preservice teacher using scenario method. *SAGE Open*, 15(2). <https://doi-org/10.1177/21582440251344753>

Baek, S. G., **Kim, J**(corresponding author), & Park, H. 2021. Development and validation of the academic resilience scale for undergraduate students using scenario method. *The SNU Journal of Education Research*, 30(3), 29-57.

IN PRESS

Kim, J., & Thoemmes, F. J. Systematic Literature Review for Mediation Analysis *Collabra: Psychology*

- ▶ Systematically analyze current trends of conducting mediation analysis in social psychology
- ▶ Suggest appropriate ways for mediation analysis
- ▶ Preregistration and materials available at <https://osf.io/dtk4h/overview>

UNDER REVIEW

Kim, J., & Thoemmes, F. J. A Causal Approach to Careless Responding

- ▶ Reframe careless responding as a causal problem rather than treating it solely as a form of measurement error

Thoemmes, F. J. & Kim, J. The Roles of Machine Learning and Large Language Models in Causal Inference

- ▶ Examined the potential uses of machine learning and large language models across three stages of causal inference: design, analysis, and auditing.

Kim, J. & Kim, Y. Statistical Evidence of Sex Discrimination Using Instrumental Inequality

- ▶ Defined the notion of discrimination using DAG
- ▶ Proved whether there is statistical evidence of discrimination utilizing instrumental variable on a discrimination case

Soleimani, A. D., Kim, J. & Sternberg, R. J. An Augmented Triangular Theory of the Expression of Love

- ▶ Using LPA, distinguished different love expression latent profiles

IN PROGRESS

Kim, J. Partial Identification with Careless Responding

- ▶ Propose a causal partial-identification approach to careless responding, deriving bounds for target estimands under alternative assumptions about how careless responses are generated.

Kim, J., & Thoemmes, F. J. To Remove or Not to Remove? The Consequences of Careless Responding in Psychological Data

- ▶ Provide simulation-based guidance on when careless responses should be eliminated, conditional on the causal structure of careless responding

Kim, J., & Thoemmes, F. J. Graphical Criteria of Recoverability under Not Missing at Random Case

- ▶ Distinguish subtypes of missing-not-at-random according to their bias occurrence using a causal graphical model

Kim, J., & Thoemmes, F. J. Causal Graphical Representation of Difference-in-Discontinuity

- ▶ Developing a graphical representation of the difference-in-discontinuities design to clarify the causal assumptions required for valid application

Kim, J., & Thoemmes, F. J. Causal Assumptions for Two-Wave Longitudinal Mediation Analysis

- ▶ Confirm existing assumptions for two-wave longitudinal mediation using causal graphical model

Kim, J., & Thoemmes, F. J. On the Importance of Taking Causal Assumptions in Developmental Psychopathology Seriously.

- ▶ **Invited contribution** to a special issue of *Development and Psychopathology*. Draft submitted.

Thoemmes, F. J., & Kim, J. Review of *Causal Mediation Analysis* by Geoffrey T. Wodtke and Xiang Zhou.

- ▶ **Invited book review** for *The American Statistician*. In preparation.

Presentations

* *presenting author*

ORAL PRESENTATIONS

***Kim, J., & Kim, Y. 2026. Credible Evidence of Sex Discrimination Using Instrumental Inequality (Forthcoming).** Center for the Study of Inequality. Ithaca, U.S.

***Kim, J., & Thoemmes, F. 2026. Causal Approach for Careless Responding .** Symposia Presentation: International Meeting of the Psychometric Society (IMPS). Seoul, South Korea.

Kim, J., & *Thoemmes, F. 2026. A Systematic Literature Review of Causal Mediation Analysis in Social Psychology. Oral Presentation: EASP General Meeting 2026. Strasbourg, France.

- ***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: Michigan State University. East Lansing, U.S.
- ***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: International Conference of Educational Research (ICER). Seoul, South Korea
- ***Kim, J.**, & Thoemmes, F. 2025. Systematic Literature Review for Mediation Analysis . Oral Presentation: Brown Bags. Ithaca, U.S.
- ***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: 2025 International Conference on Education Research (ICER). Seoul, South Korea
- Kim, J.** , Baek, J., & *Lee, H. 2025. Development of the Pre-service Teacher Competency Scale in an Online Learning Environment. Oral Presentation: American Educational Research Association (AERA), Denver, U.S.
- *Kim, Y., & **Kim, J.** 2022. Statistical Evidence for Inequality: Applying Causal Restriction of Instrumental Variable. Oral Presentation: Korea Inequality Research Network Conference, Seoul, Korea.

POSTER PRESENTATIONS

- ***Kim, J.**, & Thoemmes, F. 2026. Revisiting Sequential Ignorability in the Presence of Missing Data (Forthcoming). Poster: 2026 The Society of Multivariate Experimental Psychology (SMEP). Durango, U.S.
- ***Kim, J.**, & Thoemmes, F. 2026. Causal Approach for Careless Responding . Poster: 2026 American Causal Inference Conference (ACIC). Salt Lake City, U.S.
- ***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Poster: 2025 American Causal Inference Conference (ACIC). Detroit, U.S.
- ***Kim, J.**, & Thoemmes, F. 2024. Causal Assumptions of the Two-Wave Longitudinal Mediation Model . Poster: 2024 The Society of Multivariate Experimental Psychology (SMEP). Ithaca, U.S.
- ***Kim, J.**, & Kim, Y. 2024. Credible Evidence of Gender Discrimination Using Instrumental Inequality. Poster: 2024 American Causal Inference Conference (ACIC). Seattle, U.S.
- ***Kim, J.** 2022. Table 2 Fallacy in Latent Class Analysis. Poster: 22nd International Conference on Educational Research (ICER). Seoul, Korea.
- Baek, J., ***Kim, J.**, Lee, H., Choi, Y.& Kim, S. 2022. Pre-Service Teachers' Experiences and Perceptions of Teacher Preparation Programs: How Are Teachers Educated to Teach Virtually?. Poster: 22nd International Conference on Education Research (ICER). Seoul, Korea.
- ***Kim, J.**, & Park, H. 2022. Measurement and Analysis of S University students' academic resilience. Online Poster: SNU BK21 Seminar
- ***Kim, J.**, Nam, C., & Seo, Y. 2017. A Narrative Exploration of the EBS Internet Lecture Experience of Low-Income High School Students. Poster: Korea Educational Research Society, Daejeon, Korea.

Teaching Experience

Human Bonding

Cornell University

TEACHING ASSISTANT

Spring 2026

- Supported instruction for more than 800 students in a large-enrollment undergraduate course on interpersonal relationships and human behavior.
- Responded to student questions, facilitated discussions, and provided guidance on course readings and assignments.

Statistics and Research Design

Cornell University

TEACHING ASSISTANT

Fall 2025

- Supported students in implementing and interpreting analyses in R, connecting statistical output to the underlying research question.
- Assisted students in understanding basic statistics, including NHST and the Bayesian approach.

Statistics and Research Design

Cornell University

GUEST LECTURER

Fall 2025

- Introduced how to use directed acyclic graphs and explained confounders, mediators, and colliders and their roles.

Quantitative Methods-2

Cornell University

GUEST LECTURER

Spring 2025

- Introduced the counterfactual framework for causal mediation analysis, including the definitions and interpretation of natural direct, natural indirect, and controlled direct effects.

Causal Mediation Analysis

GESIS Workshop Series

TEACHING ASSISTANT

Fall 2024

- Supported participants in implementing causal mediation analyses in R and connecting the code to the underlying estimands and assumptions.
- Provided targeted methodological references in response to participants' questions, enabling further study beyond the workshop.
- Participant feedback described my support as "very active and helpful" and "very attentive," specifically noting the provision of relevant references in real time.

Multiple Regression Course (Center for Teaching and Learning)

Seoul National University

TEACHING ASSISTANT

Spring 2021

- Guided students in implementing and interpreting multiple regression models in R.

Moral Education

Sahmyook Middle School

PRACTICING TEACHER

Spring 2019

- Designed flipped-learning activities that used class time for discussion and application of moral and ethical reasoning.
- Provided individualized guidance to middle-school students during discussion-based activities.

Mathematics

Samsung Dream Class

AFTERSCHOOL TEACHER

Spring 2017 – Fall 2018

- Taught mathematics to underprivileged middle school students.
- Delivered a special lecture on college life and career pathways.

Korean for North Korean Defectors

Yeomyung School

AFTERSCHOOL TEACHER

Spring 2018

- Prepared students for the Korean fluency test.
- Advised a delinquent student as a homeroom teacher.
- Organized a career lecture featuring professionals from various fields.

Professional Experience

The Korea Chamber of Commerce and Industry

Seoul, South Korea

SENIOR RESEARCHER

Aug. 2022 – Mar. 2023

- Developed National Competency Standard test items for vocational high school students.
- Applied the Rasch model to analyze test results.
- Equated tests using the bookmark method.
- Wrote policy reports on vocational education and student competency evaluation.
- Collaborated with the Ministry of Education on test development.
- Revised tests to reflect changes in the national curriculum.

Korea Institute for Research in the Behavioral Sciences

Seoul, South Korea

RESEARCHER

Jan. 2022 – Jul. 2022

- Developed test items to assess competency and morality.
- Revised and updated norms for psychological assessments.
- Provided individualized feedback to test takers.
- Analyzed personality profiles using the CPI test scale.
- Assessed predictive validity and reported results to client firms.

Brain Korea 21, Department of Education, Seoul National University

Seoul, South Korea

RESEARCHER

Sep. 2021 – Jan. 2022

- Assessed academic resilience among undergraduate students using psychometric scales.
- Supported development and validation of new measurement tools.
- Participated in educational methodology workshops.

Awards, Fellowships, & Grants

2026	The Society of Multivariate Experimental Psychology (SMEP) , Travel grant	\$2,000
2026	Woowa Small Business Owners' Children Scholarship , Woowa Brothers	\$10,000
2026	Psychometric Society Travel Award (IMPS) , ACS Ventures sponsorship	\$1,000
2024	Appenzeller Memorial Fellowship , Ewha Foundation	\$3,000
2023	Sage Fellowship , Cornell University	\$64,186
2021	Scholarship , Korea Association of University Women Excellence RA Award , Seoul National University Gwanak Residence Hall	\$1,000

Mentoring

Front-Door Project

Cornell University

RESEARCH MENTOR

Jan 2026 – August 2026

- Mentored an undergraduate researcher in computer science department in extending the front-door criterion to more general causal structures through targeted readings and guidance on graphical identification.
- Co-developed generalizable R code for identifying front-door structures and submitted the implementation as a pull request to the developers of the **dagitty** package.

Harvard Bio Summer Student Research Program

Harvard University

RESEARCH MENTOR

June 2024 – August 2024

- Helped a student translate an interest in medicine and antidepressant use into a researchable causal question using an instrumental-variable framework.
- Guided the student in identifying candidate variables and evaluating possible violations of the required identification assumptions.

Gwanak Residence Hall

Seoul National University

RESIDENT ADVISOR

June 2020 – December 2021

- Advised undergraduate students on adjusting to university life.
- Planned and organized residence hall festivals.
- Provided assistance to students during COVID-19 isolation.
- Supported international students facing psychological challenges.

Dream School for Children of Traffic Accident Victims

Hyundai Motor

DREAM MENTOR

Spring 2016 – Fall 2016

- Designed individualized career development programs.
- Mentored a child of traffic accident victims to overcome academic challenges.
- Facilitated opportunities for students to meet inspiring role models.

Outreach & Professional Development

STATISTICAL LANGUAGE

► R, SPSS, Amos, $\text{\LaTeX}$

LANGUAGE

► Native: Korean

► Fluent: English

LICENSURE

► Teacher's Certificate for the Secondary School Teacher of Moral Education, Ministry of Education, Republic of Korea.

SERVICE

- ▶ Reviewer - CRC Press, Causal Inference Textbook (2026)
- ▶ Reviewer - Journal of Causal Inference (2026)
- ▶ Reviewer - Psychological Methods (2025, 2026)
- ▶ Volunteer - ACIC (American Causal Inference Conference) (2025, 2026)
- ▶ Reviewer - Multivariate Behavioral Research (2025)
- ▶ Reviewer - Scientific Studies of Readings (2025)
- ▶ Reviewer - Psychological Test Adaptation and Development (2025)
- ▶ Student Council in Ewha Womans University (2016-2018)
- ▶ Leader of Educational Academic Society in Ewha Womans University (2016-2017)